

Introduction

Remark 1.1. This document considers stochastic order notation, limits of sets, different modes of stochastic convergence: almost sure, in L^p , in probability, and in distribution; and uniform integrability. Throughout, assume that $(\mathbf{X}_n)$ is a sequence of scalar or vector-valued random variables with candidate limit $\mathbf{X}$, and that the $(\mathbf{X}_n)$ and $\mathbf{X}$ are defined on an underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$. ◆

Order Notation

2.1 Definitions

Definition 2.1. Let α_n and β_n denote sequences of real numbers, then $\alpha_n = \mathcal{O}(\beta_n)$ if there exist a bound $M \in \mathbb{R}^+$ and a threshold $\nu \in \mathbb{N}$ s.t. for $n \geq \nu$: $|\alpha_n| \leq M|\beta_n|$. ■

Definition 2.2. A sequence of random variables $(\mathbf{X}_n)$ is **bounded in probability**, expressed $\mathbf{X}_n = \mathcal{O}_p(1)$, if for $\forall \epsilon > 0$ there $\exists (M_\epsilon, \nu_\epsilon)$ s.t. $n \geq \nu_\epsilon$ implies:

$$P(\|\mathbf{X}_n\| > M_\epsilon) < \epsilon.$$

If the sequence of random variables $(\mathbf{X}_n)$ is *bounded in probability*, then the corresponding sequence (F_n) of probability measures is described as *uniformly tight*. ■

Definition 2.3. Let α_n and β_n denote sequences of real numbers, then $\alpha_n = o(\beta_n)$ if for $\forall \epsilon > 0$ there $\exists (\nu_\epsilon)$ s.t. for $n \geq \nu_\epsilon$: $|\alpha_n| \leq \epsilon|\beta_n|$. ■

Definition 2.4. A sequence of random variables $(\mathbf{X}_n)$ **converges in probability** to zero, expressed $\mathbf{X}_n = o_p(1)$, if for $\forall \epsilon > 0$:

$$\lim_{n \rightarrow \infty} \mathbb{P}(\|\mathbf{X}_n\| > \epsilon) = 0.$$

Convergence of the sequence $(\mathbf{X}_n)$ in probability to zero requires that for $\forall \epsilon, \delta > 0$ there exists ν_δ s.t. when $n \geq \nu_\delta$ the probability $P(\|\mathbf{X}_n\| > \epsilon) < \delta$. ■

Definition 2.5. Suppose $(\mathbf{X}_n)$ is a sequence of random variables, and that $(\alpha_n) \in \mathbb{R}^+$ is a sequence of positive constants.

- i. $\mathbf{X}_n = o_p(\alpha_n) \iff \alpha_n^{-1} \mathbf{X}_n = o_p(1)$.
 - ii. $\mathbf{X}_n = \mathcal{O}_p(\alpha_n) \iff \alpha_n^{-1} \mathbf{X}_n = \mathcal{O}_p(1)$.
-

2.2 Properties

Proposition 2.1. If $\mathbf{X}_n$ converges in probability to zero, then $\mathbf{X}_n$ is bounded in probability: $\mathbf{X}_n = o_p(1) \implies \mathbf{X}_n = \mathcal{O}_p(1)$. ◆

Proof. Fix $\epsilon > 0$, then by the definition of convergence in probability, for $\forall \delta > 0$ there $\exists \nu_\delta \in \mathbb{N}$ s.t. when $n \geq \nu_\delta$, $\mathbb{P}(\|\mathbf{X}_n\| > \epsilon) < \delta$. ■

Proposition 2.2 (Sub-additivity). Suppose $\{\mathbf{X}_i\}$ is a finite collection of random variables, not necessarily independent nor identically distributed. Then:

$$\mathbb{P}\left(\left\|\sum_{i=1}^n \mathbf{X}_i\right\|_2 > \epsilon\right) \leq \sum_{i=1}^n \mathbb{P}(\|\mathbf{X}_i\| > \epsilon/n) \quad (2.1)$$
◆

Proof. If $\left\|\sum_{i=1}^n \mathbf{X}_i\right\|_2 > \epsilon$, then at least one $\|\mathbf{X}_i\| > \epsilon/n$, for suppose not, then:

$$\left\|\sum_{i=1}^n \mathbf{X}_i\right\|_2 \leq \sum_{i=1}^n \|\mathbf{X}_i\| \leq \sum_{i=1}^n \frac{\epsilon}{n} = \epsilon,$$

which leads to a contradiction. Expressed in terms of events:

$$\left\{\omega \in \Omega : \left\|\sum_{i=1}^n \mathbf{X}_i\right\|_2 > \epsilon\right\} \subset \bigcup_{i=1}^n \left\{\omega \in \Omega : \|\mathbf{X}_i\| > \epsilon/n\right\}.$$

By sub-additivity of the probability measure:

$$\mathbb{P}\left(\left\|\sum_{i=1}^n \mathbf{X}_i\right\|_2 > \epsilon\right) \leq \mathbb{P}\left(\bigcup_{i=1}^n \left\{\omega \in \Omega : \|\mathbf{X}_i\| > \epsilon/n\right\}\right) \leq \sum_{i=1}^n \mathbb{P}(\|\mathbf{X}_i\| > \epsilon/n).$$
■

Proposition 2.3. If $X_n = \mathcal{O}_p(1)$ and $Y_n = \mathcal{O}_p(1)$, then:

- i. $X_n + Y_n = \mathcal{O}_p(1)$.
 - ii. $X_n Y_n = \mathcal{O}_p(1)$.
- ◆

Proof. i. Since $X_n = \mathcal{O}_p(1)$ and $Y_n = \mathcal{O}_p(1)$. Fix $\epsilon > 0$, there $\exists(M_X, \nu_X)$ and $\exists(M_Y, \nu_Y)$ s.t. when $n \geq \nu_X$, $\mathbb{P}(\|X_n\| > M_X) < \epsilon/2$ and when $n \geq \nu_Y$, $\mathbb{P}(\|Y_n\| > M_Y) < \epsilon/2$. Set $M = \max(M_X, M_Y)$, then:

$$\mathbb{P}(\|X_n + Y_n\| > 2M) \leq \mathbb{P}(\|X_n\| > M) + \mathbb{P}(\|Y_n\| > M) \leq \epsilon/2 + \epsilon/2 = \epsilon.$$

ii.

$$\begin{aligned}\mathbb{P}(\|X_n Y_n\| > M_X M_Y) &\leq \mathbb{P}(\|X_n\| > M_X \cup \|Y_n\| > M_Y) \\ &\leq \mathbb{P}(\|X_n\| > M_X) + \mathbb{P}(\|Y_n\| > M_Y) \leq \epsilon/2 + \epsilon/2 = \epsilon.\end{aligned}$$

■

Proposition 2.4. If $X_n = o_p(1)$ and $Y_n = \mathcal{O}_p(1)$, then:

i. $X_n + Y_n = \mathcal{O}_p(1)$.

ii. $X_n Y_n = o_p(1)$.

◆

Proof. i. Since $X_n = o_p(1) \implies X_n = \mathcal{O}_p(1)$, (i.) follows from the last proposition.

ii. Fix $\epsilon > 0$. For $\forall(M, \delta)$,

$$\begin{aligned}\mathbb{P}(\|X_n Y_n\| > M) &= \mathbb{P}(\|X_n Y_n\| > \delta \cap \|Y_n\| > M) + \mathbb{P}(\|X_n Y_n\| \leq \delta \cap \|Y_n\| \leq M) \\ &\leq \mathbb{P}(\|Y_n\| > M) + \mathbb{P}(\|X_n\| > \delta/M)\end{aligned}$$

Since $Y_n = \mathcal{O}_p(1)$, for any $\epsilon > 0$ there $\exists M_\epsilon$ s.t. when $n \geq \nu_\epsilon$, $\mathbb{P}(\|Y_n\| > M_\epsilon) < \epsilon$. Moreover, since $X_n = o_p(1)$, $\lim_{n \rightarrow \infty} \mathbb{P}(\|X_n\| > \delta/M) = 0$. Thus:

$$\lim_{n \rightarrow \infty} \mathbb{P}(\|X_n Y_n\| > M_\epsilon) \leq \epsilon + \lim_{n \rightarrow \infty} \mathbb{P}(\|X_n\| > \delta/M) = \epsilon.$$

Since $\epsilon > 0$ was arbitrary, conclude $X_n Y_n = o_p(1)$.

■

Theorem 2.1. Suppose $\mathbf{X}_n = o_p(\alpha_n)$ and $\mathbf{Y}_n = o_p(\beta_n)$, then:

i. $\mathbf{X}_n + \mathbf{Y}_n = o_p\{\max(\alpha_n, \beta_n)\}$.

ii. $\mathbf{X}_n \mathbf{Y}_n = o_p(\alpha_n \beta_n)$.

iii. $\|\mathbf{X}_n\|^r = o_p(\alpha_n^r)$ where $r > 0$.

□

Proof. i. If $\|\mathbf{X}_n + \mathbf{Y}_n\| / \max(\alpha_n, \beta_n) > \epsilon$, then either:

$$\frac{\|\mathbf{X}_n\|}{\alpha_n} > \frac{\epsilon}{2} \quad \vee \quad \frac{\|\mathbf{Y}_n\|}{\beta_n} > \frac{\epsilon}{2}.$$

By subadditivity:

$$\lim_{n \rightarrow \infty} \mathbb{P}\left\{\frac{\|\mathbf{X}_n + \mathbf{Y}_n\|}{\max(\alpha_n, \beta_n)} > \epsilon\right\} \leq \lim_{n \rightarrow \infty} \mathbb{P}\left(\alpha_n^{-1} \|\mathbf{X}_n\| > \frac{\epsilon}{2}\right) + \mathbb{P}\left(\beta_n^{-1} \|\mathbf{Y}_n\| > \frac{\epsilon}{2}\right) = 0.$$

ii. If $\|\mathbf{X}_n \mathbf{Y}_n\|/(\alpha_n \beta_n) > \epsilon$, then either:

$$\{\alpha_n^{-1} \|\mathbf{X}_n\| \leq 1, \beta_n^{-1} \|\mathbf{Y}_n\| > \epsilon\} \cup \{\alpha_n^{-1} \|\mathbf{X}_n\| > 1, (\alpha_n \beta_n)^{-1} \|\mathbf{X}_n \mathbf{Y}_n\| > \epsilon\}.$$

By subadditivity:

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{\|\mathbf{X}_n \mathbf{Y}_n\|}{\alpha_n \beta_n} > \epsilon \right) \leq \lim_{n \rightarrow \infty} \mathbb{P} (\beta_n^{-1} \|\mathbf{Y}_n\| > \epsilon) + \mathbb{P} (\alpha_n^{-1} \|\mathbf{X}_n\| > 1) = 0.$$

iii.

$$\lim_{n \rightarrow \infty} \mathbb{P} (\alpha_n^{-r} \|\mathbf{X}_n\|^r > \epsilon) = \lim_{n \rightarrow \infty} \mathbb{P} (\alpha_n^{-1} \|\mathbf{X}_n\| > \epsilon^{1/r}) = 0.$$

■

Proposition 2.5.

$$\mathbf{X}_n - \mathbf{X} = o_p(1) \iff \|\mathbf{X}_n - \mathbf{X}\| = o_p(1).$$

◆

Proof. Let $\mathbf{Y}_n = \mathbf{X}_n - \mathbf{X}$, then by definition $\mathbf{Y}_n = o_p(1)$ if and only if for $\forall \epsilon > 0$:

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbb{P}(\|\mathbf{Y}_n\| > \epsilon) &= 0, \\ \lim_{n \rightarrow \infty} \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon) &= 0. \end{aligned}$$

Now let $Y_n = \|\mathbf{X}_n - \mathbf{X}\|$, then $Y_n = o_p(1)$ if and only if for $\forall \epsilon > 0$:

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbb{P}(|Y_n| > \epsilon) &= 0, \\ \lim_{n \rightarrow \infty} \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon) &= 0. \end{aligned}$$

Thus, the statements $\mathbf{X}_n - \mathbf{X} = o_p(1)$ and $\|\mathbf{X}_n - \mathbf{X}\| = o_p(1)$ are identical. ■

Proposition 2.6. Suppose $(\mathbf{X}_n)$ and $(\mathbf{Y}_n)$ are sequences of random variables. If 1. $\mathbf{X}_n - \mathbf{Y}_n = o_p(1)$ and 2. $\mathbf{Y}_n - \mathbf{Y} = o_p(1)$, then $\mathbf{X}_n - \mathbf{Y} = o_p(1)$. ◆

Proof. By the triangle inequality:

$$\|\mathbf{X}_n - \mathbf{Y}\| \leq \|\mathbf{X}_n - \mathbf{Y}_n\|_2 + \|\mathbf{Y}_n - \mathbf{Y}\| = o_p(1)$$

■

Proposition 2.7. Suppose $(\mathbf{X}_n)$ is a sequence of J dimension random variables, and $(\alpha_n) \in \mathbb{R}^+$ is a sequence of positive constants, then:

i. $\mathbf{X}_n = o_p(1) \iff X_{nj} = o_p(1)$ for $j \in \{1, \dots, J\}$.

ii. $\mathbf{X}_n = \mathcal{O}_p(1) \iff X_{nj} = \mathcal{O}_p(1)$ for $j \in \{1, \dots, J\}$.

That is, a sequence of random variables converges in probability to zero, or is bounded in probability, if and only if the components convergence in probability to zero, or are bounded in probability. $\blacklozenge$

Proof. i. ($\implies$):

$$|X_{nj} - X_j| \leq \sqrt{\sum_{j=1}^k (X_{nj} - X_j)^2},$$

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_{nj} - X_j| > \epsilon) \leq \lim_{n \rightarrow \infty} \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon) = 0.$$

($\impliedby$):

$$\lim_{n \rightarrow \infty} \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon) \leq \lim_{n \rightarrow \infty} \sum_{j=1}^J \mathbb{P}(|X_{nj} - X_j| > \epsilon/J) = 0.$$

ii. ($\implies$) Since $\mathbf{X}_n = \mathcal{O}_p(1)$, for $\forall \epsilon > 0$ there $\exists (M_\epsilon, \nu_\epsilon)$ s.t. when $n \geq \nu_\epsilon$:

$$\mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > M_\epsilon) < \epsilon.$$

Since $\mathbb{P}(|X_{nj} - X_j| > M_\epsilon) \leq \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > M_\epsilon)$, conclude $X_{nj} = \mathcal{O}_p(1)$.

For $\epsilon > 0$ there $\exists M_\epsilon, \nu_\epsilon$ s.t. when $n \geq \nu_\epsilon$:

$$\mathbb{P}(|X_{nj} - X_j| > M_\epsilon) \leq \mathbb{P}\left(\sqrt{\sum_{j=1}^k (X_{nj} - X_j)^2} > M_\epsilon\right) = \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > M_\epsilon) < \epsilon.$$

($\impliedby$) Fix $\epsilon > 0$. For each $j \in \{1, \dots, J\}$ there $\exists M_j, \nu_j$ s.t. when $n \geq \nu_j$:

$$\mathbb{P}(|X_{nj} - X_j| > M_j) < \frac{\epsilon}{J}.$$

Set $M = \max_j M_j$ and $\nu = \max_j \nu_j$. Now when $n \geq \nu$:

$$\mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > JM) \leq \sum_{j=1}^J \mathbb{P}(|X_{nj} - X_j| > M) \leq J \cdot \frac{\epsilon}{J} = \epsilon.$$

$\blacksquare$

Limits of Sets

3.1 Definitions

Definition 3.1. Suppose (B_n) is a *decreasing* sequence of measurable sets. The limit $\lim_{n \rightarrow \infty} B_n$ is defined as their intersection:

$$\lim_{n \rightarrow \infty} B_n = \bigcap_{n=1}^{\infty} B_n.$$

$\omega \in \lim_{n \rightarrow \infty} B_n$ if for $\forall n \in \mathbb{N}$, $\omega \in B_n$. ■

Definition 3.2. Suppose (C_n) is an *increasing* sequence of measurable sets. The limit $\lim_{n \rightarrow \infty} C_n$ is defined as their union:

$$\lim_{n \rightarrow \infty} C_n = \bigcup_{n=1}^{\infty} C_n.$$

$\omega \in \lim_{n \rightarrow \infty} C_n$ if there exists an $n \in \mathbb{N}$ s.t. $\omega \in C_n$. ■

Definition 3.3. Define the *supremum* of sequence of sets as:

$$\sup_{k \geq n} A_k = \bigcup_{k \geq n} A_k.$$

The **limit supremum** of a sequence (A_n) of sets is:

$$\limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k \geq n} A_k.$$

Observe that $B_n = \sup_{k \geq n} A_k$ is a *decreasing* sequence of sets, since each consecutive B_n is the union of fewer A_n . ■

Definition 3.4. Define the *infimum* of a sequence of sets as:

$$\inf_{k \geq n} A_k = \bigcap_{k \geq n} A_k.$$

The **limit infimum** of a sequence (A_n) of sets is:

$$\liminf_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k \geq n} A_k$$

Observe that $C_n = \inf_{k \geq n} A_k$ is an *increasing* sequence of sets, since each consecutive C_n is the intersection of fewer A_n . ■

Definition 3.5. The limit supremum and limit infimum of a sequence of sets (A_n) always exist. If these two sets are equal, then the **limit** exists and is defined as:

$$\lim_{n \rightarrow \infty} A_n \equiv \liminf_{n \rightarrow \infty} A_n = \limsup_{n \rightarrow \infty} A_n.$$

■

3.2 Properties

Proposition 3.1. For any sequence of sets,

$$\liminf_{n \rightarrow \infty} A_n \subseteq \limsup_{n \rightarrow \infty} A_n.$$

◆

Proof. Suppose $\omega \in \liminf_{n \rightarrow \infty} A_n$, then there exists an $n \in \mathbb{N}$ s.t. $\omega \in A_k$ for $\forall k \geq n$. That is, ω belongs to every A_k far enough out in the sequence. Thus, for any $n \in \mathbb{N}$, there exists $k \geq n$ s.t. $\omega \in A_k$. Conclude that $\omega \in \limsup_{n \rightarrow \infty} A_n$. ■

Remark 3.1. Since the limit infimum is always a subset of the limit supremum, to prove the limit $\lim_{n \rightarrow \infty} A_n$ exists, it suffices to prove that the limit supremum is a subset of the limit infimum. ◆

Proposition 3.2. Suppose $C_n \rightarrow C$ is an increasing sequence of measurable sets on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, then:

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} C_n\right) = \lim_{n \rightarrow \infty} \mathbb{P}(C_n).$$

◆

Proof. Define the sequence of *disjoint* sets $D_1 = C_1$, and $D_k = C_k - C_{k-1}$ for $k \geq 2$. Clearly $C_n = \cup_{k=1}^n D_k$. By finite additivity of the probability measure:

$$\mathbb{P}(C_n) = \mathbb{P}\left(\bigcup_{k=1}^n D_k\right) = \sum_{k=1}^n \mathbb{P}(D_k).$$

Note too that since the C_n are increasing:

$$\lim_{n \rightarrow \infty} C_n = \bigcup_{n=1}^{\infty} C_n = \bigcup_{n=1}^{\infty} \bigcup_{k=1}^n D_k = \bigcup_{k=1}^{\infty} D_k.$$

By σ -additivity of the probability measure $\mathbb{P}$:

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} C_n\right) = \mathbb{P}\left(\bigcup_{k=1}^{\infty} D_k\right) = \sum_{k=1}^{\infty} \mathbb{P}(D_k) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \mathbb{P}(D_k) = \lim_{n \rightarrow \infty} \mathbb{P}(C_n).$$

■

Corollary 3.1. Suppose $B_n \rightarrow B$ is a decreasing sequence of measurable sets, then:

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} B_n\right) = \lim_{n \rightarrow \infty} \mathbb{P}(B_n).$$

♣

Proof. Since (B_n) is decreasing, the sequence of complements (B_n^c) is necessarily increasing, thus:

$$\begin{aligned}\mathbb{P}\left(\lim_{n \rightarrow \infty} B_n^c\right) &= \lim_{n \rightarrow \infty} \mathbb{P}(B_n^c), \\ 1 - \mathbb{P}\left(\lim_{n \rightarrow \infty} B_n^c\right) &= 1 - \lim_{n \rightarrow \infty} \mathbb{P}(B_n^c).\end{aligned}$$

The LHS is:

$$1 - \mathbb{P}\left(\lim_{n \rightarrow \infty} B_n^c\right) = P\left(\bigcup_{n=1}^{\infty} \bigcap_{k \geq n} B_k^c\right)^c = P\left(\bigcap_{n=1}^{\infty} \bigcup_{k \geq n} B_k\right) = \mathbb{P}\left(\lim_{n \rightarrow \infty} B_n\right).$$

The RHS is:

$$1 - \lim_{n \rightarrow \infty} \mathbb{P}(B_n^c) = \lim_{n \rightarrow \infty} \{1 - \mathbb{P}(B_n^c)\} = \lim_{n \rightarrow \infty} \mathbb{P}(B_n).$$

■

Corollary 3.2. Let (A_n) denote a sequence of sets, then:

$$\begin{aligned}\lim_{n \rightarrow \infty} \mathbb{P}\left(\sup_{k \geq n} A_k\right) &= \mathbb{P}\left(\limsup_{n \rightarrow \infty} A_n\right) \\ \lim_{n \rightarrow \infty} \mathbb{P}\left(\inf_{k \geq n} A_k\right) &= \mathbb{P}\left(\liminf_{n \rightarrow \infty} A_n\right)\end{aligned}$$

♣

Proof. The conclusion follows because $B_n = \sup_{k \geq n} A_k$ is decreasing and $C_n = \inf_{k \geq n} A_k$ is increasing. ■

Theorem 3.1 (Continuity). If (A_n) is a sequence of sets converging to A , then:

$$\lim_{n \rightarrow \infty} \mathbb{P}(A_n) = \mathbb{P}\left(\lim_{n \rightarrow \infty} A_n\right). \quad (3.1)$$

□

Proof. Since $\inf_{k \geq n} A_k \subseteq A_n \subseteq \sup_{k \geq n} A_k$:

$$\mathbb{P}\left(\inf_{k \geq n} A_k\right) \leq \mathbb{P}(A_n) \leq \mathbb{P}\left(\sup_{k \geq n} A_k\right).$$

Taking the limit as $n \rightarrow \infty$:

$$\begin{aligned}\lim_{n \rightarrow \infty} \mathbb{P}\left(\inf_{k \geq n} A_k\right) &\leq \lim_{n \rightarrow \infty} \mathbb{P}(A_n) \leq \lim_{n \rightarrow \infty} \mathbb{P}\left(\sup_{k \geq n} A_k\right), \\ \mathbb{P}\left(\liminf_{n \rightarrow \infty} A_n\right) &\leq \lim_{n \rightarrow \infty} \mathbb{P}(A_n) \leq \mathbb{P}\left(\limsup_{n \rightarrow \infty} A_n\right).\end{aligned}$$

Since $A_n \rightarrow A$:

$$\lim_{n \rightarrow \infty} A_n = \liminf_{n \rightarrow \infty} A_n = \limsup_{n \rightarrow \infty} A_n.$$

Conclude that:

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} A_n\right) \leq \lim_{n \rightarrow \infty} \mathbb{P}(A_n) \leq \mathbb{P}\left(\lim_{n \rightarrow \infty} A_n\right).$$

■

Almost Sure Convergence

Definition 4.1. A sequence of random variables $(\mathbf{X}_n)$ **converges almost surely** to $\mathbf{X}$, expressed $\mathbf{X}_n \xrightarrow{as} \mathbf{X}$, if:

$$\mathbb{P}\left\{\omega : \lim_{n \rightarrow \infty} \mathbf{X}_n(\omega) = \mathbf{X}(\omega)\right\} = 1.$$

Equivalently, for $\forall \epsilon > 0$:

$$\mathbb{P}\left\{\omega : \limsup_{n \rightarrow \infty} \|\mathbf{X}_n - \mathbf{X}\| > \epsilon\right\} = 0.$$

■

Remark 4.1. In the following, the notation:

$$\left\{\limsup_{n \rightarrow \infty} \|\mathbf{X}_n - \mathbf{X}\| > \epsilon\right\},$$

will implicitly refer to the set of $\omega \in \Omega$ where the condition $\{\cdot\}$ holds.

◆

4.1 Criteria

Proposition 4.1. If for $\forall \epsilon > 0$:

$$\sum_{n=1}^{\infty} \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon) < \infty,$$

then $\mathbf{X}_n \xrightarrow{as} \mathbf{X}$.

◆

Proof.

$$\begin{aligned} \mathbb{P}\left\{\limsup_{n \rightarrow \infty} \|\mathbf{X}_n - \mathbf{X}\| > \epsilon\right\} &= \lim_{n \rightarrow \infty} \mathbb{P}\left\{\sup_{k \geq n} \|\mathbf{X}_k - \mathbf{X}\| > \epsilon\right\} \\ &= \lim_{n \rightarrow \infty} \mathbb{P}\left\{\bigcup_{k \geq n} \|\mathbf{X}_k - \mathbf{X}\| > \epsilon\right\} \leq \lim_{n \rightarrow \infty} \sum_{k=n}^{\infty} \mathbb{P}(\|\mathbf{X}_k - \mathbf{X}\| > \epsilon). \end{aligned}$$

Since the series $\sum_{n=1}^{\infty} \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\|_2 > \epsilon)$ converges,

$$\lim_{n \rightarrow \infty} \sum_{k=n}^{\infty} \mathbb{P}(\|\mathbf{X}_k - \mathbf{X}\| > \epsilon) = 0.$$

■

Proposition 4.2. If $\exists(p > 0)$ such that:

$$\sum_{n=1}^{\infty} \mathbb{E}\|\mathbf{X}_n - \mathbf{X}\|^p < \infty,$$

then $\mathbf{X}_n \xrightarrow{as} \mathbf{X}$.

◆

Proof.

$$\begin{aligned} \mathbb{P}\left\{\limsup_{n \rightarrow \infty} \|\mathbf{X}_n - \mathbf{X}\| > \epsilon\right\} &\leq \lim_{n \rightarrow \infty} \sum_{k=n}^{\infty} \mathbb{P}(\|\mathbf{X}_k - \mathbf{X}\| > \epsilon) \\ &\leq \frac{1}{\epsilon^p} \lim_{n \rightarrow \infty} \sum_{k=n}^{\infty} \mathbb{E}\|\mathbf{X}_k - \mathbf{X}\|^p = 0. \end{aligned}$$

■

4.2 Relation to Expectation

Remark 4.2. The following theorems carry over from the analogous results for sequences of real valued functions since if $\mathbf{X}_n \xrightarrow{as} \mathbf{X}$, then $\lim_{n \rightarrow \infty} \mathbf{X}_n(\omega) = \mathbf{X}(\omega)$, except possibly on a set of measure zero.

◆

Theorem 4.1 (Monotone Convergence). If (X_n) is a non-negative, increasing sequence of scalar random variables with $X_n \xrightarrow{as} X$, then:

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = \mathbb{E}(X).$$

□

Theorem 4.2 (Fatou's). If (X_n) is a non-negative sequence of random variables, with $X_n \xrightarrow{as} X$, then:

$$\mathbb{E}(X) = \mathbb{E}\left(\liminf_{n \rightarrow \infty} X_n\right) \leq \liminf_{n \rightarrow \infty} \mathbb{E}(X_n).$$

□

Theorem 4.3 (Dominated Convergence). If $X_n \xrightarrow{as} X$ and $\|\mathbf{X}_n\| \leq Y$ with $\mathbb{E}(Y) < \infty$, then:

$$\lim_{n \rightarrow \infty} \mathbb{E}(\mathbf{X}_n) = \mathbb{E}(\mathbf{X}).$$

□

4.3 Relation to Convergence in Probability

Proposition 4.3. Almost sure convergence implies convergence in probability:

$$\mathbf{X}_n \xrightarrow{as} \mathbf{X} \implies \mathbf{X}_n \xrightarrow{p} \mathbf{X}.$$

◆

Proof. Suppose $\mathbf{X}_n \xrightarrow{as} \mathbf{X}$, then for $\forall \epsilon > 0$:

$$\begin{aligned} 0 &= \mathbb{P} \left\{ \limsup_{n \rightarrow \infty} \|\mathbf{X}_n - \mathbf{X}\| > \epsilon \right\} \\ &= \lim_{n \rightarrow \infty} \mathbb{P} \left\{ \sup_{k \geq n} \|\mathbf{X}_k - \mathbf{X}\| > \epsilon \right\} \\ &\geq \lim_{n \rightarrow \infty} \mathbb{P} \{ \|\mathbf{X}_n - \mathbf{X}\| > \epsilon \}. \end{aligned}$$

■

L^p Convergence

Definition 5.1. A sequence of random variables $(\mathbf{X}_n)$ **converges in L^p** , expressed $\mathbf{X}_n \xrightarrow{L^p} \mathbf{X}$ if:

$$\lim_{n \rightarrow \infty} \mathbb{E}(\|\mathbf{X}_n - \mathbf{X}\|^p) = 0.$$

■

Proposition 5.1 (Markov's Inequality).

$$\mathbb{P}(\|\mathbf{X}\| \geq t) \leq \frac{\mathbb{E}\|\mathbf{X}\|}{t}.$$

◆

Proof. Let $Y = \|\mathbf{X}\|$,

$$\mathbb{P}(Y \geq t) = \mathbb{E}\{\mathbb{I}(Y \geq t)\} = \mathbb{E}\{\mathbb{I}(Y/t \geq 1)\} \leq \mathbb{E}(Y/t).$$

■

Corollary 5.1. For $p > 0$,

$$\mathbb{P}(\|\mathbf{X}\| \geq t) \leq \frac{\mathbb{E}\|\mathbf{X}\|^p}{t^p}.$$

♣

Proposition 5.2. If $\mathbf{X}_n \xrightarrow{L^p} \mathbf{X}$ then:

$$\lim_{n \rightarrow \infty} \mathbb{E} \|\mathbf{X}_n\|^p = \lim_{n \rightarrow \infty} \mathbb{E} \|\mathbf{X}\|^p.$$

◆

Proof. By Minkowski's inequality in L^p : $(\mathbb{E} \|\mathbf{X}_n - \mathbf{X}\|^p)^{1/p} \leq (\mathbb{E} \|\mathbf{X}_n\|^p)^{1/p} + (\mathbb{E} \|\mathbf{X}\|^p)^{1/p}$, and $(\mathbb{E} \|\mathbf{X}\|^p)^{1/p} \leq (\mathbb{E} \|\mathbf{X}_n - \mathbf{X}\|^p)^{1/p} + (\mathbb{E} \|\mathbf{X}_n\|^p)^{1/p}$. Thus:

$$|(\mathbb{E} \|\mathbf{X}_n\|^p)^{1/p} - (\mathbb{E} \|\mathbf{X}\|^p)^{1/p}| \leq (\mathbb{E} \|\mathbf{X}_n - \mathbf{X}\|^p)^{1/p} \rightarrow 0.$$

Since the map $u \mapsto u^p$ is continuous and $\mathbf{X}_n \xrightarrow{L^p} \mathbf{X}$ implies $\mathbb{E} \|\mathbf{X}_n - \mathbf{X}\|^p \rightarrow 0$, conclude $\lim_{n \rightarrow \infty} \mathbb{E} \|\mathbf{X}_n\|^p = \mathbb{E} \|\mathbf{X}\|^p$. ■

5.1 Relation to Convergence in Probability

Proposition 5.3. For $p > 1$, convergence in L^p implies convergence in probability:

$$\mathbf{X}_n \xrightarrow{L^p} \mathbf{X} \implies \mathbf{X}_n \xrightarrow{p} \mathbf{X}.$$

◆

Proof. For $\forall \epsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon) \leq \frac{1}{\epsilon^p} \lim_{n \rightarrow \infty} \mathbb{E} \|\mathbf{X}_n - \mathbf{X}\|^p = 0.$$

■

Convergence in Probability

Definition 6.1. A sequence of random variables $(\mathbf{X}_n)$ **converges in probability** to $\mathbf{X}$, expressed $\mathbf{X}_n \xrightarrow{p} \mathbf{X}$, if for $\forall \epsilon > 0$:

$$\lim_{n \rightarrow \infty} \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon) = 0.$$

■

Definition 6.2. A sequence of random variables $(\mathbf{X}_n)$ is **Cauchy in probability** if $\forall \epsilon, \delta > 0$ there exists $\nu_{\epsilon, \delta}$ such that when $n, m \geq \nu_{\epsilon, \delta}$:

$$\mathbb{P}(\|\mathbf{X}_n - \mathbf{X}_m\| > \epsilon) < \delta.$$

■

6.1 Relation to Almost Sure Convergence

Proposition 6.1. If $\mathbf{X}_n \xrightarrow{p} \mathbf{X}$, then $(\mathbf{X}_n)$ is Cauchy in probability. ◆

Proof. By sub-additivity (2.1):

$$\mathbb{P}(\|\mathbf{X}_n - \mathbf{X}_m\| > \epsilon) \leq \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon/2) + \mathbb{P}(\|\mathbf{X} - \mathbf{X}_m\| > \epsilon/2).$$

Since $\mathbf{X}_n \xrightarrow{p} \mathbf{X}$, there $\exists \nu_{\epsilon, \delta}$ such that when $n \geq \nu_{\epsilon, \delta}$:

$$\mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon/2) < \delta/2$$

Now for $n, m \geq \nu_{\epsilon, \delta}$:

$$\mathbb{P}(\|\mathbf{X}_n - \mathbf{X}_m\| > \epsilon) < \delta/2 + \delta/2 = \delta.$$

■

Proposition 6.2. If $(\mathbf{X}_n)$ is Cauchy in probability, then $(\mathbf{X}_n)$ contains a subsequence $(\mathbf{X}_{n_j})$ that converges almost surely. ◆

Proof. Let $n_1 = 1$, and define n_j by:

$$n_j = \inf_{n \in \mathbb{N}} \{n > n_{j-1} : \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}_m\| > 2^{-j}) < 2^{-j} \text{ for } \forall (n, m) \geq n\}.$$

Thus, n_j is the smallest n exceeding n_{j-1} such that the probability of the distance between $\mathbf{X}_n$ and $\mathbf{X}_m$ exceeding $\epsilon = 2^{-j}$ is less than $\delta = 2^{-j}$. Now since:

$$\sum_{j=1}^{\infty} \mathbb{P}(\|\mathbf{X}_{n_{j+1}} - \mathbf{X}_{n_j}\| > 2^{-j}) < \infty,$$

by the Borel-Cantelli lemmas:

$$\mathbb{P} \left\{ \limsup_{j \rightarrow \infty} (\|\mathbf{X}_{n_{j+1}} - \mathbf{X}_{n_j}\| > 2^{-j}) \right\} = 0$$

almost everywhere on Ω . Hence for ω in a set of probability one:

$$\alpha_j(\omega) = \|\mathbf{X}_{n_{j+1}}(\omega) - \mathbf{X}_{n_j}(\omega)\| \leq 2^{-j}.$$

Now $\{\alpha_j(\omega)\}$ is a Cauchy sequence in $\mathbb{R}$, and by completeness, $\alpha_j(\omega)$ converges to some limit $\alpha(\omega)$. Thus, $\lim_{j \rightarrow \infty} \mathbf{X}_{n_j}(\omega)$ exists almost everywhere on Ω . ■

Proposition 6.3. If $(\mathbf{X}_n)$ is Cauchy in probability, then $\mathbf{X}_n$ converges in probability to some limit $\mathbf{X}$. ◆

Proof. Since $(\mathbf{X}_n)$ is Cauchy in probability, there exists a subsequence $(\mathbf{X}_{n_j})$ converging almost surely to some limit $\mathbf{X}$. Now:

$$\mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon) \leq \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}_{n_j}\| > \epsilon/2) + \mathbb{P}(\|\mathbf{X}_{n_j} - \mathbf{X}\| > \epsilon/2).$$

Since $(\mathbf{X}_n)$ is Cauchy, there exists ν_1 such that when $n, n_j \geq \nu_1$:

$$\mathbb{P}(\|\mathbf{X}_n - \mathbf{X}_{n_j}\| > \epsilon/2) < \delta/2.$$

Since $\mathbf{X}_{n_j} \xrightarrow{as} \mathbf{X}$, which implies $\mathbf{X}_{n_j} \xrightarrow{p} \mathbf{X}$, there exists ν_2 such that when $n_j \geq \nu_2$:

$$\mathbb{P}(\|\mathbf{X}_{n_j} - \mathbf{X}\| > \epsilon/2) < \delta/2.$$

Let $\nu = \max(\nu_1, \nu_2)$, then for $n, n_j \geq \nu$:

$$\mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon) < \delta.$$

■

Proposition 6.4. $\mathbf{X}_n \xrightarrow{p} \mathbf{X}$ if and only if every subsequence $(\mathbf{X}_{n_j})$ itself contains a subsequence $(\mathbf{X}_{n_{jk}})$ converging almost surely to $\mathbf{X}$. ♦

Proof. ($\implies$) If $\mathbf{X}_n \xrightarrow{p} \mathbf{X}$, then any subsequence $(\mathbf{X}_{n_j})$ converges in probability to the same limit, and the subsequence $(\mathbf{X}_{n_j})$ contains a further subsequence converging almost surely.

($\impliedby$) Suppose for contradiction that every subsequence contains an almost surely converging subsequence, but that $\mathbf{X}_n$ does not converge in probability. Since it is not the case that $\mathbf{X}_n \xrightarrow{p} \mathbf{X}$, for $\delta > 0$ and $\epsilon > 0$ there exist infinitely many k such that:

$$\mathbb{P}(\|\mathbf{X}_k - \mathbf{X}\| > \epsilon) \geq \delta.$$

Let $\mathbf{X}_{n_j}$ denote a subsequence along such k , then $(\mathbf{X}_{n_j})$ can itself contain no subsequence converging almost surely to $\mathbf{X}$. ■

Theorem 6.1 (Cauchy Criteria).

- i. $(\mathbf{X}_n)$ converges in probability if and only if $(\mathbf{X}_n)$ is Cauchy in probability.
- ii. $\mathbf{X}_n \xrightarrow{p} \mathbf{X}$ if and only if every subsequence $(\mathbf{X}_{n_j})$ contains a further subsequence $(\mathbf{X}_{n_{jk}})$ converging almost surely to $\mathbf{X}$.

□

6.2 Relation to Expectation

Theorem 6.2 (Dominated Convergence). If $\mathbf{X}_n \xrightarrow{p} \mathbf{X}$ and $\|\mathbf{X}_n\| \leq Y$ with $\mathbb{E}(Y) < \infty$, then:

$$\lim_{n \rightarrow \infty} \mathbb{E}(\mathbf{X}_n) = \mathbb{E}(\mathbf{X}).$$

□

Remark 6.1. The result is established by showing that every convergent subsequence of $\mathbb{E}(\mathbf{X}_n)$ converges to the same limit. Suppose $\mathbb{E}(\mathbf{X}_{n_j})$ is a convergent subsequence. Since $\mathbf{X}_{n_j} \xrightarrow{p} \mathbf{X}$, there exists a subsequence $\mathbf{X}_{n_{j_k}}$ converging almost surely to $\mathbf{X}$. By the dominated convergence theorem for almost sure convergence, $\mathbb{E}(\mathbf{X}_{n_{j_k}}) \rightarrow \mathbb{E}(\mathbf{X})$. Since this holds for any subsequence, every convergent subsequence of $\mathbb{E}(\mathbf{X}_n)$ approaches the same limit. See Resnick (2014), corollary 6.3.2. ♦

6.3 Uniform Integrability

Definition 6.3. Let $\mathcal{T}$ denote an index set. The collection of random variables $(\mathbf{X}_t)$ is **uniformly integrable** if:

$$\lim_{M \rightarrow \infty} \sup_{t \in \mathcal{T}} \mathbb{E}\{\|\mathbf{X}_t\| \mathbb{I}(\|\mathbf{X}_t\| > M)\} = 0.$$

■

Remark 6.2. If $(\mathbf{X}_t)$ is uniformly integrable, then for any $\epsilon > 0$, there exists M_ϵ such that for $m \geq M_\epsilon$:

$$\sup_{t \in \mathcal{T}} \mathbb{E}\{\|\mathbf{X}_t\| \mathbb{I}(\|\mathbf{X}_t\| > m)\} < \epsilon.$$

♦

Proposition 6.5. A random variable $\mathbf{X}$ has finite expectation $\mathbb{E}\|\mathbf{X}\| < \infty \iff$

$$\lim_{M \rightarrow \infty} \mathbb{E}\{\|\mathbf{X}\| \mathbb{I}(\|\mathbf{X}\| > M)\} = 0.$$

That is, an individual $\mathbf{X}$ has finite expectation if and only if it is uniformly integrable. ♦

Proof. ($\implies$) Consider the sequence of functions:

$$g_M(\omega) = \|\mathbf{X}(\omega)\| \mathbb{I}(\|\mathbf{X}(\omega)\| \leq M)$$

for $M \rightarrow \infty$. This sequence is non-negative and increasing in M , with:

$$\lim_{M \rightarrow \infty} g_M(\omega) = \|\mathbf{X}(\omega)\|.$$

By the monotone convergence theorem:

$$\lim_{M \rightarrow \infty} \mathbb{E}\{\|\mathbf{X}\| \mathbb{I}(\|\mathbf{X}\| \leq M)\} = \mathbb{E}\{\|\mathbf{X}\|\}.$$

Since:

$$\mathbb{E}\{\|\mathbf{X}\| \mathbb{I}(\|\mathbf{X}\| > M)\} = \mathbb{E}\|\mathbf{X}\| - \mathbb{E}\{\|\mathbf{X}\| \mathbb{I}(\|\mathbf{X}\| \leq M)\},$$

upon taking the limit as $M \rightarrow \infty$:

$$\lim_{M \rightarrow \infty} \mathbb{E}\{\|\mathbf{X}\| \mathbb{I}(\|\mathbf{X}\| > M)\} = \mathbb{E}\|\mathbf{X}\| - \lim_{M \rightarrow \infty} \mathbb{E}\{\|\mathbf{X}\| \mathbb{I}(\|\mathbf{X}\| \leq M)\} = 0.$$

($\Leftarrow$) Suppose on the contrary that $\mathbb{E}\|\mathbf{X}\| = \infty$. Since:

$$\mathbb{E}\{\|\mathbf{X}\| \mathbb{I}(\|\mathbf{X}\| \leq M)\} \leq M \mathbb{P}(\|\mathbf{X}\| \leq M) \leq M,$$

and:

$$\mathbb{E}\|\mathbf{X}\| = \mathbb{E}\{\|\mathbf{X}\| \mathbb{I}(\|\mathbf{X}\| \leq M)\} + \mathbb{E}\{\|\mathbf{X}\| \mathbb{I}(\|\mathbf{X}\| > M)\},$$

if $\mathbb{E}\|\mathbf{X}\| = \infty$, then for any M :

$$\mathbb{E}\{\|\mathbf{X}\| \mathbb{I}(\|\mathbf{X}\| > M)\} = \infty.$$

Thus $\lim_{M \rightarrow \infty} \mathbb{E}\{\|\mathbf{X}\| \mathbb{I}(\|\mathbf{X}\| > M)\} \neq 0$. ■

Proposition 6.6. If $(\mathbf{X}_t)$ is uniformly integrable, if and only if:

- i. $\sup_{t \in \mathcal{T}} \mathbb{E}\|\mathbf{X}_t\| < \infty$,
- ii. For $\forall \epsilon > 0$ there $\exists \delta > 0$ s.t.

$$\mathbb{P}(A) < \delta \implies \sup_{t \in \mathcal{T}} \mathbb{E}\{\|\mathbf{X}_t\| \mathbb{I}_A\} < \epsilon.$$

Importantly, $\sup_{t \in \mathcal{T}} \mathbb{E}\|\mathbf{X}_t\| < \infty$ is necessary but not sufficient for a sequence $(\mathbf{X}_t)$ to achieve uniform integrability. ◆

Proof. ($\implies$) Suppose $(\mathbf{X}_t)$ is uniformly integrable. Fix $\epsilon > 0$. Choose M_ϵ such that $\sup_{t \in \mathcal{T}} \mathbb{E}\{\|\mathbf{X}_t\| \mathbb{I}(\|\mathbf{X}_t\| > M_\epsilon)\} < \epsilon$. Now for $\forall t$:

$$\mathbb{E}\|\mathbf{X}_t\| = \mathbb{E}\{\|\mathbf{X}_t\| \mathbb{I}(\|\mathbf{X}_t\| \leq M_\epsilon)\} + \mathbb{E}\{\|\mathbf{X}_t\| \mathbb{I}(\|\mathbf{X}_t\| > M_\epsilon)\} \leq M_\epsilon + \epsilon.$$

Thus $\sup_{t \in \mathcal{T}} \mathbb{E}\|\mathbf{X}_t\| < \infty$. Now select M_ϵ such that $\sup_{t \in \mathcal{T}} \mathbb{E}\{\|\mathbf{X}_t\| \mathbb{I}(\|\mathbf{X}_t\| > M_\epsilon)\} < \epsilon/2$ and set $\delta = \epsilon/(2M_\epsilon)$. If $\mathbb{P}(A) < \delta$, then:

$$\begin{aligned} \mathbb{E}\{\|\mathbf{X}_t\| \mathbb{I}_A\} &= \mathbb{E}\{\|\mathbf{X}_t\| \mathbb{I}(A \cap \|\mathbf{X}_t\| \leq M_\epsilon)\} + \mathbb{E}\{\|\mathbf{X}_t\| \mathbb{I}(A \cap \|\mathbf{X}_t\| > M_\epsilon)\} \\ &\leq M_\epsilon \mathbb{P}(A) + \mathbb{E}\{\|\mathbf{X}_t\| \mathbb{I}(\|\mathbf{X}_t\| > M_\epsilon)\} \leq M_\epsilon \frac{\epsilon}{2M_\epsilon} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

($\Leftarrow$) By condition (i.), there exists K such that $\sup_{t \in \mathcal{T}} \mathbb{E} \|\mathbf{X}_t\| < K$. By condition (ii.), for $\epsilon > 0$, there exists $\delta > 0$ s.t.

$$\mathbb{P}(A) < \delta \implies \sup_{t \in \mathcal{T}} \mathbb{E} \{ \|\mathbf{X}_t\| \mathbb{I}_A \} < \epsilon.$$

Observe that by Markov's inequality:

$$\mathbb{P}(\|\mathbf{X}_i\| > M) \leq \frac{\mathbb{E}(\|\mathbf{X}_i\|)}{M} \leq \frac{K}{M}.$$

Choose M such that $K/M < \delta$, then $\mathbb{P}(\|\mathbf{X}_i\| > M) < \delta$, which implies that:

$$\sup_{t \in \mathcal{T}} \mathbb{E} \{ \|\mathbf{X}_t\| \mathbb{I}(\|\mathbf{X}_i\| > M) \} < \epsilon,$$

as required. ■

Proposition 6.7. If there exists $\delta > 0$ such that:

$$\sup_{t \in \mathcal{T}} \mathbb{E} \|\mathbf{X}_t\|^{1+\delta} < \infty,$$

then $(\mathbf{X}_t)$ is uniformly integrable. ◆

Proof.

$$\begin{aligned} \mathbb{E} \{ \|\mathbf{X}_t\| \mathbb{I}(\|\mathbf{X}_t\| > M) \} &= \mathbb{E} \{ \|\mathbf{X}_t\| \cdot 1 \cdot \mathbb{I}(\|\mathbf{X}_t\|/M > 1) \} \\ &\leq \mathbb{E} \left\{ \|\mathbf{X}_t\| \cdot \frac{\|\mathbf{X}_t\|^\delta}{M^\delta} \cdot \mathbb{I}(\|\mathbf{X}_t\|/M > 1) \right\} \\ &\leq \frac{1}{M^\delta} \mathbb{E} \{ \|\mathbf{X}_t\|^{1+\delta} \} \rightarrow 0 \end{aligned}$$

as $M \rightarrow \infty$. ■

Theorem 6.3. The following statements are equivalent:

- i. $\mathbf{X}_n \xrightarrow{L^1} \mathbf{X}$,
- ii. $(\mathbf{X}_n)$ is uniformly integrable, and $\mathbf{X}_n \xrightarrow{p} \mathbf{X}$.

□

Proof. ($\Leftarrow$) Since $\mathbf{X}_n \xrightarrow{p} \mathbf{X}$, there exists a subsequence $(\mathbf{X}_{n_k})$ converging almost surely to $\mathbf{X}$. By Fatou's lemma:

$$\mathbb{E} \{ \|\mathbf{X}\| \} = \mathbb{E} \left\{ \liminf_{n_k \rightarrow \infty} \|\mathbf{X}_{n_k}\| \right\} \leq \liminf_{n_k \rightarrow \infty} \mathbb{E} \|\mathbf{X}_{n_k}\| \leq \sup_{n \in \mathbb{N}} \mathbb{E} \|\mathbf{X}_n\| < \infty,$$

where the last inequality follows from uniform integrability. Define the bounded, continuous function:

$$\psi_M(t) = \begin{cases} -M, & t \leq -M, \\ t, & t \in (-M, M), \\ M, & t \geq M. \end{cases}$$

Consider:

$$\mathbb{E}||\mathbf{X}_n - \mathbf{X}|| \leq \mathbb{E}||\mathbf{X}_n - \psi_M(\mathbf{X}_n)|| + \mathbb{E}||\psi_M(\mathbf{X}_n) - \psi_M(\mathbf{X})|| + \mathbb{E}||\psi_M(\mathbf{X}) - \mathbf{X}||.$$

The first term is bounded as:

$$\mathbb{E}||\mathbf{X}_n - \psi_M(\mathbf{X}_n)|| \leq \mathbb{E}\{||\mathbf{X}_n||\mathbb{I}(|\mathbf{X}_n| > M)\}.$$

By uniform integrability, there exists M_1 such that for $m \geq M_1$:

$$\sup_{n \in \mathbb{N}} \mathbb{E}\{||\mathbf{X}_n||\mathbb{I}(|\mathbf{X}_n| > m)\} < \frac{\epsilon}{3}.$$

Since $\psi_M(t)$ is a bounded, continuous function, and $\mathbf{X}_n \xrightarrow{d} \mathbf{X}$, by the portmanteau theorem there exists ν such that for $n \geq \nu$:

$$\mathbb{E}||\psi_M(\mathbf{X}_n) - \psi_M(\mathbf{X})|| < \frac{\epsilon}{3}.$$

Finally, since $\mathbb{E}||\mathbf{X}||$ is bounded, there exists M_2 such that for $m \geq M_2$:

$$\mathbb{E}||\psi_M(\mathbf{X}) - \mathbf{X}|| \leq \mathbb{E}\{||\mathbf{X}||\mathbb{I}(|\mathbf{X}| > M)\} < \frac{\epsilon}{3}.$$

Take $n \geq \nu$ and $m \geq \max(M_1, M_2)$, then:

$$\mathbb{E}||\mathbf{X}_n - \mathbf{X}|| < \epsilon.$$

($\implies$) Since $\mathbf{X}_n \xrightarrow{L^1} \mathbf{X}$, the sequence of real numbers $\mathbb{E}||\mathbf{X}_n||$ converges to $\mathbb{E}||\mathbf{X}|| < \infty$. Since convergent sequences in $\mathbb{R}$ are bounded:

$$\sup_{n \in \mathbb{N}} \mathbb{E}||\mathbf{X}_n|| < \infty.$$

Now since $\mathbb{E}||\mathbf{X}_n - \mathbf{X}|| \rightarrow 0$, there exists ν_ϵ such that for $n \geq \nu_\epsilon$:

$$\mathbb{E}||\mathbf{X}_n - \mathbf{X}|| < \epsilon/2.$$

Let $K_1 = \mathbb{E}||\mathbf{X}||$ and $K_2 = \max_{n < \nu} \mathbb{E}||\mathbf{X}_n||$. Set $K = \max(K_1, K_2)$. Choose δ_1 such that if $\mathbb{P}(A) < \delta_1$, then:

$$\mathbb{E}\{||\mathbf{X}||\mathbb{I}_A\} \leq K\mathbb{P}(A) < \frac{\epsilon}{2}.$$

Choose δ_2 such that if $\mathbb{P}(A) < \delta_1$, then:

$$\max_{n < \nu} \mathbb{E}\{\|\mathbf{X}_n - \mathbf{X}\|_{\mathbb{I}_A}\} \leq K\mathbb{P}(A) < \frac{\epsilon}{2}$$

Set $\delta = \min(\delta_1, \delta_2)$, then for $\mathbb{P}(A) < \delta$:

$$\sup_{n \in \mathbb{N}} \mathbb{E}\{\|\mathbf{X}_n\|_{\mathbb{I}_A}\} \leq \sup_{n \in \mathbb{N}} \mathbb{E}\{\|\mathbf{X}_n - \mathbf{X}\|_{\mathbb{I}_A}\} + \mathbb{E}\{\|\mathbf{X}\|_{\mathbb{I}_A}\} < \frac{\epsilon}{2} + \frac{\epsilon}{2} < \epsilon.$$

The inequality holds for $n < \nu$ by choice of δ , and for $n \geq \nu$ by convergence in L^1 . ■

6.4 Relation to Convergence in Distribution

Proposition 6.8. Convergence in probability implies convergence in distribution:

$$\mathbf{X}_n \xrightarrow{p} \mathbf{X} \implies \mathbf{X}_n \xrightarrow{d} \mathbf{X}.$$

◆

Proof. Suppose $\mathbf{X}_n \xrightarrow{p} \mathbf{X}$ and that $\mathbf{t}$ is a continuity point of F_X , then:

$$\begin{aligned} \mathbb{P}(\mathbf{X}_n \leq \mathbf{t}) &= \mathbb{P}\left(\{\mathbf{X}_n \leq \mathbf{t}, \|\mathbf{X}_n - \mathbf{X}\| < \epsilon\mathbf{1}\} \cup \{\mathbf{X}_n \leq \mathbf{t}, \|\mathbf{X}_n - \mathbf{X}\| \geq \epsilon\mathbf{1}\}\right) \\ &\leq \mathbb{P}(\mathbf{X} \leq \mathbf{t} + \epsilon\mathbf{1}) + \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon). \end{aligned}$$

Similarly:

$$\mathbb{P}(\mathbf{X} \leq \mathbf{t} - \epsilon\mathbf{1}) \leq \mathbb{P}(\mathbf{X}_n \leq \mathbf{t}) + \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon).$$

Thus:

$$\mathbb{P}(\mathbf{X} \leq \mathbf{t} - \epsilon\mathbf{1}) - \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon) \leq \mathbb{P}(\mathbf{X}_n \leq \mathbf{t}) \leq \mathbb{P}(\mathbf{X} \leq \mathbf{t} + \epsilon\mathbf{1}) + \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon).$$

Taking the limit as $n \rightarrow \infty$:

$$\mathbb{P}(\mathbf{X} \leq \mathbf{t} - \epsilon\mathbf{1}) \leq \lim_{n \rightarrow \infty} \mathbb{P}(\mathbf{X}_n \leq \mathbf{t}) \leq \mathbb{P}(\mathbf{X} \leq \mathbf{t} + \epsilon\mathbf{1}).$$

Since $\mathbf{t}$ is a continuity point of F_X as $\epsilon \rightarrow 0$:

$$\mathbb{P}(\mathbf{X} \leq \mathbf{t}) \leq \lim_{n \rightarrow \infty} \mathbb{P}(\mathbf{X}_n \leq \mathbf{t}) \leq \mathbb{P}(\mathbf{X} \leq \mathbf{t}).$$

■

Convergence in Distribution

Definition 7.1. A sequence of random variables $(\mathbf{X}_n)$ **converges in distribution** if the sequence of distribution functions (F_n) converges *pointwise* to the distribution function F_X of $\mathbf{X}$ on the set $\mathcal{C}(F_X)$ of continuity points of F_X :

$$\lim_{n \rightarrow \infty} F_n(\mathbf{t}) = F_X(\mathbf{t}) \text{ for } \mathbf{t} \in \mathcal{C}(F_X)$$

■

7.1 Relation to Convergence in Probability

Proposition 7.1. Convergence in distribution to a constant α implies convergence in probability:

$$\mathbf{X}_n \xrightarrow{d} \alpha \implies \mathbf{X}_n \xrightarrow{p} \alpha.$$

◆

Proof. If $\mathbf{X}_n \xrightarrow{d} \alpha$, then:

$$\lim_{n \rightarrow \infty} \mathbb{P}(\mathbf{X}_n \leq t) = \mathbb{I}(t \geq \alpha).$$

Now, probability that $\|\mathbf{X}_n - \alpha\| < \epsilon$ is:

$$\mathbb{P}(\|\mathbf{X}_n - \alpha\| < \epsilon) = \mathbb{P}(\alpha - \epsilon \mathbf{1} \leq \mathbf{X}_n \leq \alpha + \epsilon \mathbf{1}) = F_n(\alpha + \epsilon \mathbf{1}) - F_n(\alpha - \epsilon \mathbf{1}).$$

Taking the limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \mathbb{P}(\|\mathbf{X}_n - \alpha\| < \epsilon) = \mathbb{I}(\alpha + \epsilon \mathbf{1} > \alpha) - \mathbb{I}(\alpha - \epsilon \mathbf{1} > \alpha) = 1.$$

■

7.2 Characteristic Functions

Definition 7.2. The **characteristic function** $\phi : \mathbb{R} \rightarrow \mathbb{C}$ of a random variable $\mathbf{X}$ with distribution F is defined by:

$$\phi(t) = \mathbb{E}(e^{itX}) = \int e^{itx} dF(x).$$

■

Theorem 7.1 (*Uniqueness*). The characteristic function $\phi(t)$ of a random variable uniquely determines its distribution. □

Remark 7.1. See Resnick (2014), theorem 9.5.1. ◆

Theorem 7.2 (**Levy Continuity**). Suppose $(\mathbf{X}_n)$ is a sequence of random variables, and $\phi_n(t)$ the corresponding sequence of characteristic functions. $\mathbf{X}_n$ converges in distribution to a random variable $\mathbf{X}$ with characteristic function $\phi(t)$ if and only if $\phi_n(t) \rightarrow \phi(t)$:

$$\mathbf{X}_n \xrightarrow{d} \mathbf{X} \iff \phi_n \rightarrow \phi.$$

Moreover, if (ϕ_n) converges *pointwise* to ϕ_∞ , and ϕ_∞ is continuous at zero, then ϕ_∞ is the characteristic function of some random variable $\mathbf{X}$ and $\mathbf{X}_n \xrightarrow{d} \mathbf{X}$. □

Remark 7.2. See Resnick (2014), theorem 9.5.2; van der Vaart (1998), theorem 2.13. ◆

7.3 Additional Properties

Theorem 7.3 (Portmanteau). The following statements are equivalent:

- i. $X_n \xrightarrow{d} X$,
- ii. $\mathbb{E}\{g(X_n)\} \rightarrow \mathbb{E}\{g(X)\}$ for every bounded, continuous function g .
- iii. $\mathbb{E}\{\mathcal{L}(X_n)\} \rightarrow \mathbb{E}\{\mathcal{L}(X)\}$ for every bounded, Lipschitz function $\mathcal{L}$.

□

Remark 7.3. For proof, see van der Vaart (1998), lemma 2.2. ♦

Theorem 7.4 (Skorokhod Representation). If $X_n \xrightarrow{d} X$, then there exists a sequence of random variables (ξ_n) defined on a common probability space such that $X_n \stackrel{d}{=} \xi_n$, $X \stackrel{d}{=} \xi$, and $\xi_n \xrightarrow{as} \xi$. □

Remark 7.4. The proof is by construction. See Billingsley (1995), theorem 25.6. ♦

Theorem 7.5 (Dominated Convergence). If $X_n \xrightarrow{d} X$ and $\|X_n\| \leq Y$ with $\mathbb{E}(Y) < \infty$, then:

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = \mathbb{E}(X).$$

□

Proof. By the Skorokhod representation theorem, there exist a sequence of random variables (ξ_n) and ξ such that $\xi_n \stackrel{d}{=} X_n$, $\xi \stackrel{d}{=} X$, and $\xi_n \xrightarrow{as} \xi$. By the dominated convergence theorem for almost sure convergence, $\mathbb{E}(\xi_n) \rightarrow \mathbb{E}(\xi)$. Thus:

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = \lim_{n \rightarrow \infty} \mathbb{E}(\xi_n) = \mathbb{E}(\xi) = \mathbb{E}(X).$$

■

Summary of Convergence Relations

- Convergence almost surely implies convergence in probability:

$$\mathbf{X}_n \xrightarrow{as} \mathbf{X} \implies \mathbf{X}_n \xrightarrow{p} \mathbf{X}.$$

- Convergence almost surely may be established by checking:

- The series $\sum_{n=1}^{\infty} \mathbb{P}(\|\mathbf{X}_n - \mathbf{X}\| > \epsilon)$ is finite for $\forall \epsilon > 0$.
- The series $\sum_{n=1}^{\infty} \mathbb{E}\|\mathbf{X}_n - \mathbf{X}\|^p$ is finite for some $p > 0$.

- Convergence in L^p implies convergence in probability:

$$\mathbf{X}_n \xrightarrow{L^p} \mathbf{X} \implies \mathbf{X}_n \xrightarrow{p} \mathbf{X}.$$

- Convergence in probability and uniform integrability implies convergence in L^1 .
- Convergence in probability implies convergence in distribution:

$$\mathbf{X}_n \xrightarrow{p} \mathbf{X} \implies \mathbf{X}_n \xrightarrow{d} \mathbf{X}.$$

- Convergence in distribution to a constant α implies convergence in probability to that constant:

$$\mathbf{X}_n \xrightarrow{d} \alpha \implies \mathbf{X}_n \xrightarrow{p} \alpha.$$

- (Dominated convergence) Suppose $\mathbf{X}_n \rightarrow \mathbf{X}$ almost surely, in probability, or in distribution. If $\|\mathbf{X}_n\| \leq Y$ and $\mathbb{E}(Y) < \infty$, then:

$$\lim_{n \rightarrow \infty} \mathbb{E}(\mathbf{X}_n) = \mathbb{E}(\mathbf{X}).$$